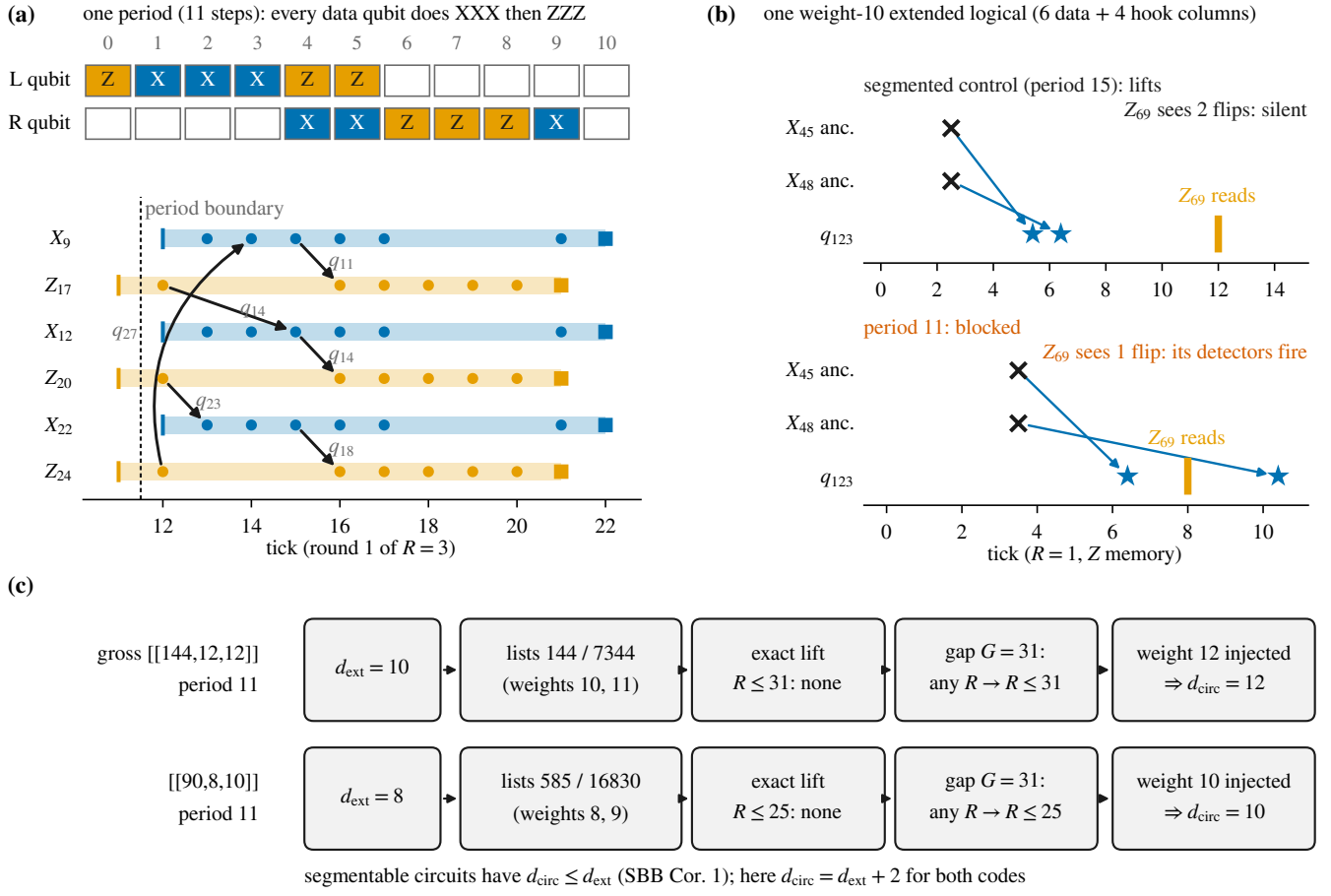
(a)
one period (11 steps): every data qubit does XXX then ZZZ
0 1 2 3 4 5 6 7 8 9 10
L qubit
R qubit
Z X X X Z Z
X X Z Z Z X
period boundary
X_9
Z_{17}
X_{12}
Z_{20}
X_{22}
Z_{24}
q_{27}
q_{11}
q_{14}
q_{14}
q_{23}
q_{18}
12 14 16 18 20 22
tick (round 1 of $R = 3$)
(b)
one weight-10 extended logical (6 data + 4 hook columns)
segmented control (period 15): lifts
Z_{69} sees 2 flips: silent
X_{45} anc.
X_{48} anc.
q_{123}
Z_{69} reads
0 2 4 6 8 10 12 14
period 11: blocked
Z_{69} sees 1 flip: its detectors fire
X_{45} anc.
X_{48} anc.
q_{123}
Z_{69} reads
0 2 4 6 8 10
tick ($R = 1$, Z memory)
(c)
gross [[144,12,12]]
period 11
$d_{ext} = 10$
lists 144 / 7344
(weights 10, 11)
exact lift
$R \leq 31$: none
gap $G = 31$:
any $R \to R \leq 31$
weight 12 injected
$\Rightarrow d_{circ} = 12$
[[90,8,10]]
period 11
$d_{ext} = 8$
lists 585 / 16830
(weights 8, 9)
exact lift
$R \leq 25$: none
gap $G = 31$:
any $R \to R \leq 25$
weight 10 injected
$\Rightarrow d_{circ} = 10$
segmentable circuits have $d_{circ} \leq d_{ext}$ (SBB Cor. 1); here $d_{circ} = d_{ext} + 2$ for both codes